Data Synthesis for Design Research—A Case Study of a Multimodal Mind Mapping Technique for Concept Generation

Varun Sai Koruprolu[1(✉)] and Gourab Kar[2]

[1] National Institute of Design, Ahmedabad, India
varunworkbin@gmail.com
[2] Indian Institute of Technology Delhi, New Delhi, India

Abstract. Design synthesis and sensemaking are crucial steps in the design process. However, translating raw data from contextual inquiry into design concepts is still a fine art. The textual approach to data synthesis is based on semantic thinking—which refers to the ability of the mind to process language in order to invoke concepts and conceptual representations. In contrast, methods such as mind mapping are based on visual thinking—which refers to the ability of the mind to analyse visual inputs and draw conclusions. While semantic and visual thinking approaches offer unique ways to uncover nuances and complexities of human behaviour germane to design research—there is a dearth of data synthesis techniques which combine the best of both approaches. This paper explores a novel and intuitive method of data synthesis that combines semantic and visual thinking approaches to seamlessly blend research insights into contextually relevant design concepts. Using a three-step process of multimodal mind mapping, researchers documented the raw data collected through observations and interviews, derived insights from the coded textual and visual data, and then translated key insights into design concepts. The feasibility of the method was demonstrated through a design project to co-create tangible interactions with and for students-with-visual-impairments. Future work includes refining the multimodal mind mapping method and exploring its versatility and suitability in different domains of design research.

Keywords: Mind Mapping · Data Synthesis · Design Concepts · Sensemaking · Design Research Methods

1 Introduction

Data synthesis is a crucial step that links contextual inquiry to concept ideation in the design process. Contextual inquiry helps designers immerse in the natural habitat of users, enabling firsthand observation of how people interact with products or services. Design researchers immerse themselves in the rhythm of the user's life, understanding not only what they do but also why they do it. However, synthesising raw data from contextual inquiry into design concepts can be challenging. Traditionally, qualitative data analysis methods are used to synthesise data from observations, interviews, field

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notes etc., into analytical text memos that are coded and categorised [1]. In contrast, methods such as mind mapping are based on visual thinking—the ability of the mind to analyse visual inputs and draw conclusions [2, 5, 6]. While semantic and visual thinking [3, 4] approaches offer unique ways to uncover nuances and complexities of human behaviour, there is a dearth of data synthesis techniques [7] which combine the best of both approaches. Therefore, this study advocates for a novel and intuitive approach to data synthesis that combines semantic thinking and visual thinking approaches to translate research insights into contextually relevant design concepts. Research questions seek to uncover: (a) How does one represent observations visually and spatially for creative synthesis? (b) How does one translate textual information into a format suitable for mind mapping?

2 Methodology

A design research study at a school for students-with-visual-impairments (SVI) provided context for generating the research data. Researchers employed shadowing, observations, and interviews with students to explore their daily actions, social interactions and non-visual sensory experiences. Over a two-week period, researchers observed 100 students, teachers, parents and other relevant stakeholders in the school. This was followed up by in-depth interviews with fifteen stakeholders. Once the research data was collected, a novel multimodal mind mapping method was used to make sense of the data.

2.1 Data Types & Mediums

Field notes served as the foundation for the data collection, capturing immediate observations and noteworthy points during fieldwork. These notes provided a firsthand account of the researcher's experiences and insights [1]. Complementing textual documentation, photography and videography were utilised to visually record significant observations, enabling a comprehensive understanding of the research context. Audio and video recordings of conversations with stakeholders and interviews preserved valuable insights and allowed for detailed analysis in the later phases of the project. Online sources, including websites, Wikipedia, academic papers, and books, served as repositories of background information and contextualised the research findings within existing literature. These varied media converged to form a comprehensive dataset, essential for synthesis and ideation of research outcomes [1].

2.2 Steps to Data Synthesis Using Multimodal Mind Mapping

Step 1: Information Collection. Initially, researchers were tasked with collecting information in any form or media from multiple sources. This comprehensive information gathering phase allowed for a wide array of data to be accumulated.

Step 2: Reflective Re-examination. Next, researchers rechecked, reflected, revisited, and re-examined all data accumulated in Step 1. By approaching the information with a fresh, unbiased perspective, researchers were able to critically question previously

unchallenged assumptions and strived to extract design insights. This reflective stage, best undertaken after completing data collection, required a dedicated timeframe equal to or in some cases greater than the information collection phase. All insights and findings from this process were compiled into a text document, while image collections remained separate.

Step 3: Sensemaking Through Mind Mapping. Finally, insights from Step 2 were used in the process of sensemaking. Researchers created colour-coded mind maps to: (a) visually represent observations; (b) identify insights, key themes and design concepts; (c) trace design concepts to their original data sources. These mind maps were displayed on a large wall which ensured that the work was constantly visible to researchers and facilitated the process of reflection and synthesis.

2.3 Application

Method 1. In synthesising text into mind maps, especially from interviews, the process began by placing the user's sketch at the centre of the sheet, symbolising them as the primary source of information. Themes and insights emerging from the semantic analysis of the interviews were annotated as shown below (Fig. 1.)



Fig. 1. Mind Mapping Method 1

Method 2. In the second mind mapping method, the process began with filtering relevant images from the database and identifying key moments or user behaviours in videos

by taking screenshots. Selected images and screenshots were digitally formatted in a collage grid and then printed on an A3-sized paper. Themes and insights emerging from the analysis of the visual data were annotated as shown below (Fig. 2.)



Fig. 2. Mind Mapping Method 2

2.4 Multimodal Mind Mapping Layers: From Observation to Ideation

The multimodal mind mapping is a three-step process which is described below (Table 1):

Layer 1: Visible Data: Researchers document the raw observations, data obtained through interview transcripts and data from secondary sources without any subjective interpretation. This layer, coded in blue colour, forms the foundational level of data collection—capturing initial data points and statements directly from the source.

Layer 2: Invisible Insights: Researchers add insights to the observed data while visually representing the synthesised text as a mind map. This layer, coded in red colour, builds upon the foundational level of data in Layer 1 by incorporating interpretations, insights and reflections related to the research problem or brief. This layer enhances comprehension by providing context and relevance to the data.

Layer 3: Idea Formation: Information from the textual and visual layers is translated into the generation of design concepts. This layer, coded in green colour, serves to

extrapolate concepts, or features from the first two layers, highlighting the most valuable takeaways and guiding the development of actionable design solutions.

Table 1. An example to show how visible data is translated to insights and design concepts.

Visible Data	Invisible Insights	Idea Formation
Teachers use ballpoint pens to grade the test papers and then verbally communicate the test scores to SVIs.	SVIs are unable to check mistakes on test papers due to lack of tactile feedback on the graded test papers.	Develop a grading pen for teachers that can create tactile marks to indicate correct & incorrect answers.

3 Findings

Two examples illustrate how the multimodal mind mapping technique was able to translate raw data to insights and design concepts with respect to non-visual sensory experiences of SVI:

Tactile Sketch Board. SVI were unable to access information on papers pinned up on a school notice board. Papers are printed in normal ink with no tactile features. SVI can comprehend simple 2D tactile diagrams with braille annotations, but do not understand when perspective is introduced. SVI communities have limited drawing skills, which is the foundation for visualisation. Therefore, the insight generated was: *How might we encourage the SVI to draw?* As a result of this insight, the concept generated was a low-cost tactile drawing board called the Tactosketch Board. This is a drawing board that uses a mesh fitted on cardboard. A PVI can draw with a ballpoint pen/stylus by placing any paper over the mesh board. Additionally, the research team developed stencils for more than twenty shapes which can be placed on paper and then the inner boundary can be traced and/or the area inside the stencil can be scribbled to get an embossed impression of the 2D objects. This prototype enabled SVIs the opportunity to draw, play, and express themselves, many for the first time in their lives.

Touching Cultures. SVIs need a companion to assist them in deciding what clothes to buy. Usually, SVI selects clothes based on what they hear from their friends/family/social media. While a sighted person shops for clothes based on looks whereas a SVI shops based on societal preference. SVI did not know the visual characteristics of other cultures/professions because they rarely had the opportunity to touch and feel the garments worn by other people. Therefore, the insight generated was: *How might we expand the tactile vocabulary of SVIs so that they can learn about cultures/professions/fashion trends?* As a result of this insight the concept generated was a set of mannequins dressed in different garments called Touching Cultures. The prototypes enabled SVIs to touch and experience fully dressed and groomed mannequins of male and female genders without the awkwardness of touching people of the opposite gender. The mannequins were

portrayed as policemen, celebrities, tennis players, watchmen, etc., giving the SVIs a first-time experience of understanding what most of them have imagined through verbal descriptions until now.

4 Discussion

In conclusion, the multimodal mind mapping technique of data synthesis combines semantic and visual thinking approaches to seamlessly blend insights into contextually relevant design concepts. The accuracy or practicality of the insights is not critical—the key is to generate a wide variety of design concepts. The final multimodal mind map can generate a wealth of concepts that are evaluated based on specific criteria. At any point, if the purpose of a concept needs to be revisited, the mind map serves as a quick reference to trace the designer's thinking. This method offers designers a practical and immersive way to engage with the mind mapping process, effectively translating complex interview and observational data into structured, visually dynamic maps that leads to creative thinking and innovation.

The strengths of the study include a practical demonstration of the multimodal mind mapping process in designing tangible interactions for and with SVI. The process of combining visual and textual data to derive insights and develop design concepts is the key contribution to knowledge. The limitations of the study include its application to a single research-informed-design project. A small design research team led by the first author was involved in conceptualising the research, conducting the field study and analysing the research data. Scalability of the process for larger design research teams is a factor that needs to be investigated in future.

Augmented reality can be used to overlay digital information on physical mind maps, adding an interactive layer where users can scan their paper mind maps to access additional digital content. Integrating machine learning algorithms with the multimodal mind mapping technique could significantly enhance its utility by assisting researchers in gaining qualitative insights, formulating the right questions, and sparking insights for design concepts. Finally, conducting studies to compare the effectiveness of physical mind mapping against digital methods in various contexts will provide valuable insights into when and how to best use each approach, ensuring the method's overall effectiveness and adaptability.

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